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# Health and performance of Holstein versus dairy × beef calves: A retrospective single-cohort study

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## ABSTRACT

Within the past decade, the use of beef semen in the dairy industry has risen, leading to a growing number of dairy × beef crossbred calves being born. However, limited studies compare the health and growth performance of purebred dairy and crossbred calves during the preweaning phase. This study aimed to evaluate the association of breed with morbidity, mortality, and growth performance during an 84-d rearing period. A total of 640 male calves were enrolled upon arrival to a calf-rearing facility and were classified as either Holstein (n = 446) or crossbred (n = 194), which were identified by hide color. Breed, source (local farm versus auction), and BW were recorded upon arrival. Calves were fed milk replacer (MR) twice daily until weaning, which occurred between d 42 and 63, and solid feed and water were offered ad libitum throughout the study. Health exams were performed twice daily, evaluating fecal consistency from arrival to d 21 and respiratory health from arrival to d 84. Mortalities and treatments administered, including electrolytes, nonsteroidal anti-inflammatory drugs, and antibiotics, were recorded throughout the study. Solid feed intake and BW were measured weekly following enrollment, and MR refusals were recorded daily. Statistical models, including Cox proportional hazards, Weibull, negative binomial, and linear regression, were built in Stata 18 (StataCorp LP) to assess the association of breed with calf health and growth. Holstein calves had a greater incidence of diarrhea compared with crossbred calves (incidence rate ratio = 1.23; 95% CI: 1.01–1.51) and a higher hazard of receiving a second (hazard ratio [HR] = 1.62; 95% CI: 1.04 to 2.51) or third (HR = 2.47; 95% CI: 1.24 to 4.91) treatment for respiratory disease. With respect to growth performance, crossbred calves had increased BW by d 28 and weighed 7.13 ± 1.31 kg (95% CI: 4.56 to 9.70) more than Holsteins by d 84. Similarly, ADG was greater for crossbreds starting between d 28 and 35. Feed-to-gain conversion ratios (FCR) were lower for crossbred calves during both the preweaning and weaning phases; however, Holstein calves had improved FCR between d 70 and 77 compared with crossbred calves. These results suggest that crossbred calves may be able to overcome illness quicker and with fewer therapeutic interventions than Holsteins in the rearing period while achieving superior growth performance.
**Key words:** dairy × beef, health, growth, feed conversion

## INTRODUCTION

In recent years, the number of dairy producers using sexed semen to selectively breed for replacement heifers has substantially increased, offering an opportunity to incorporate beef semen into dairy breeding programs (De Vries et al., 2008; Pereira et al., 2022). As a result, dairy × beef crossbred calves have become more prevalent in North America, offering dairy producers an alternative income source due to the higher market value of crossbred calves compared with Holsteins (Buczinski et al., 2021; Les Producteurs de Bovins du Quebec, 2024). In 2016, the average price of a week-old crossbred calf was $168 CAD ($123 USD), compared with $208 CAD ($152 USD) for Holsteins. Recent reports in 2025 indicate that the same crossbred calves now sell for an average of $1,659 CAD ($1,212 USD), which is $538 CAD ($393 USD) more than the average Holstein (McCarthy et al., 2026). These calves are typically managed in rearing systems similar to grain-fed veal production early in life, but ultimately enter the beef supply chain shortly after weaning. Compared with purebred Holstein steers, crossbreds tend to have greater ADG during the finishing period, are more feed efficient, and result in a more desirable end product (Pimentel-Concepción et al., 2024). However, comparative studies focusing on breed-specific health and growth performance are scarce, especially during the preweaning period.

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In North America, nonreplacement calves, including both Holstein and crossbreds not intended for milk production, are typically sold within the first days of life and transported to calf-rearing facilities (Felix et al., 2023). There, they are commonly comingled with calves from multiple sources, transitioned to new diets, exposed to various pathogens, castrated, and weaned at an early age (McGill and Sacco, 2020). These stressors place considerable strain on their naïve immune systems. Consequently, bovine respiratory disease (**BRD**) and gastrointestinal disorders emerge as the leading health challenges in preweaning calves (Schönecher et al., 2020; Abdallah et al., 2022). In the short term, producers face increased veterinary costs and labor demands (Kaneene and Hurd, 1990), and the long-term consequences include reduced growth performance and slaughter weight (Pardon et al., 2013; Marcato et al., 2020).

Crossbreeding may offer a valuable strategy to improve calf health by producing animals that are naturally more resilient and robust (Sørensen et al., 2008). Resilience, defined as the animal's ability to recover and return to a normal state following stressors such as disease, and robustness, the capacity to maintain productivity in the face of recurring challenges, are important traits in animal health (Colditz and Hine, 2016; Marcato et al., 2022). Although much of the research on crossbreeding has focused on growth and production traits, especially in the feedlot phase, some studies have also identified health benefits in early life that align with these concepts. For instance, crossbreeding has resulted in a lower incidence and shorter duration of diarrhea (Maltecca et al., 2006; Moroz et al., 2025). Most of these findings originate from studies on dairy × dairy crosses, and only recently has attention shifted to dairy × beef crossbred calves. In contrast, the reported incidence rate of respiratory disease for dairy × beef crossbred calves appears comparable to that of Holstein heifers (Rhodes et al., 2021; Fernandes et al., 2025). However, Fernandes et al. (2025) noted that crossbred calves demonstrated compensatory growth following respiratory disease, suggesting greater robustness and resilience. Although the preliminary findings are promising, indicating potential health advantages, evidence specific to dairy × beef crossbreds remains limited, especially in North America, and requires further investigation.

Therefore, the objective of this study was to evaluate the association of calf breed with growth performance, feed efficiency, the incidence of diarrhea and BRD, and mortality risk during the first 84 d at a calf-rearing facility. It was hypothesized that male crossbred calves would exhibit superior growth performance and feed efficiency, lower incidences of diarrhea and BRD, and a higher likelihood of survival compared with purebred male Holstein calves.

## MATERIALS AND METHODS

A retrospective cohort study was conducted at a single calf-rearing facility in southwestern Ontario, Canada, between June 2023 and January 2025. The facility was selected because of its extensive data records and willingness to participate in the study. Calves enrolled in this analysis were part of various nutritional trials, resulting in slight variations in feeding schemes across individuals. The animal use protocol was approved by the Animal Care Committee at the University of Guelph (AUP #4966).

### *Housing*

Calves were sourced from either local dairy farms within a 30-min radius of the facility or auction facilities and enrolled upon arrival, which was considered as d 0, to the calf-rearing facility (Mapleview Agri Ltd., Palmerston, ON, Canada). Calves arrived in groups of 64 to fill each of the 4 rooms of the facility every 3 wk. Only groups that contained both Holstein and crossbred calves were included in the study; therefore, 10 groups that arrived between June 2023 and October 2024 were selected. The study population included all male Holstein (n = 446; 48.2 ± 0.06 kg) and male crossbred (n = 194; 47.9 ± 0.10 kg) calves within the 10 groups. Crossbred calves were identified only by their black hide color. However, due to the prevalent use of Angus semen in the Ontario dairy industry, it may be assumed that many of the crossbreds were Holstein × Angus (Sweett and Fleming, 2024). The exact age upon arrival was not known, but the calves were estimated to be less than 10 d of age (Wilson et al., 2023). Each room was organized into 4 rows of 16 pens per row. Calves were individually housed in 107.50 × 98.50 × 195.50 cm pens with rubber slatted flooring from arrival until d 63. Following weaning (d 63 to 84), the partitions were removed between adjacent pens in 5 of the 10 groups and calves were paired for the postweaning period. In the remaining 5 groups, the partitions were not removed, and the calves remained single housed.

### *Feeding Scheme*

Calves were fed milk replacer (**MR**) twice a day (at 0545 and 1600 h) in a clean bucket with a floating teat (Kerbl Calf Bucket, Kerbl, Petersberg, Germany) for the first 21 d, and then directly from the pail until weaning. A step up-step down feeding scheme was used for all calves; however, the quantity of MR varied between and within groups and is depicted in Table 1. All MR was mixed at 130 g of MR powder per 1 L of water. Additionally, the composition of MR was different across groups, with calves receiving MR that contained either 26% protein, 20% fat, and 4.8 Mcal/kg of ME (3 groups); 26%

**Table 1.** Daily allowance of MR powder in grams of MR/d (and volume in liters) from arrival to weaning for Holstein (n = 446) and crossbred (n = 194) calves

| Day | Quantity A,[1] g (L) | Quantity B,[2] g (L) | Quantity C,[2] g (L) |
|---|---|---|---|
| 0– 6 | 728 (5.6) | 832 (6.4) | 650 (5.0) |
| 7–13 | 728 (5.6) | 832 (6.4) | 650 (5.0) |
| 14–20 | 910 (7.0) | 1,040 (8.0) | 780 (6.0) |
| 21–27 | 910 (7.0) | 1,040 (8.0) | 1,040 (8.0) |
| 28–34 | 910 (7.0) | 1,040 (8.0) | 1,040 (8.0) |
| 35–41 | 910 (7.0) | 1,040 (8.0) | 1,040 (8.0) |
| 42–48 | 650 (5.0) | 780 (6.0) | 780 (6.0) |
| 49–55 | 520 (4.0) | 650 (5.0) | 520 (4.0) |
| 56–63[3] | 260 (2.0) | 260 (2.0) | 260 (2.0) |
| Total (g) | 45,682 | 52,598 | 47,320 |

[1]Holstein and crossbred calves from 2 groups.

[2]Holstein and crossbred calves from 8 groups.

[3]During the final week of weaning, calves were fed MR only once per day.

protein, 17% fat, and 4.7 Mcal/kg of ME (2 groups); or 24% protein, 20% fat, and 4.8 Mcal/kg of ME (5 groups; Mapleview Agri Ltd., Mapleton, ON, Canada). Milk refusals were weighed (OP 900A, Optima Scale, Rancho Cucamonga, CA) and recorded daily.

In addition to MR, all calves had ad libitum access to texturized calf starter (Wallenstein Feed Supply, Wallenstein, ON, Canada) upon arrival to the rearing facility. The starter was formulated to contain 88.2% DM, 22.0% CP, 4.7% crude fat, 20.5% NDF, 12.6% ADF, 30.2% starch, and 2.81 Mcal/kg of ME, all on a DM basis. Following weaning on d 63, calves either continued to have access to the calf starter (7 groups) or they were provided a corn-blend diet consisting of 2.5 parts of calf starter to 1 part corn (3 groups). The corn blend was formulated to contain 86.8% DM, 18.4% CP, 5.0% crude fat, 10.1% NDF, 4.4% ADF, 51.2% starch, and 3.1 Mcal/kg of ME, on a DM basis. Solid feed was offered in a trough along the front of the pen from the time of arrival and intakes were recorded weekly. For the 5 groups that were housed in pairs during the postweaning period, feed intake was recorded as the total intake for the 2 calves within a pen and divided equally. Each pen was fitted with an individual flow meter (JLP, Carlon Meter Inc., MI) and nipple drinker, which was located inside the pen, to allow for ad libitum access to fresh water. The facility was designed to minimize mixing of water with the solid feed. Calves were randomly assigned to the different feeding schemes, and no breed differences were detected.

### Serum Total Protein and Health-Scoring System

To evaluate serum total protein (**STP**), blood samples were collected upon arrival via jugular venipuncture into a 10-mL sterile serum vacutainer tube (Becton Dickinson, NJ) from each calf in 8 of the 10 groups. The blood was allowed to clot at room temperature for up to 2 h and the samples were centrifuged (2,500 × *g* for 15 min at 4°C). Once the serum separated, a digital refractometer (Misco, OH) was used to evaluate STP. The STP concentrations at arrival were categorized as poor (<5.1 g/dL), fair (5.1–5.7 g/dL), good (5.8–6.1 g/dL), and excellent (≥6.2 g/dL) transfer of passive immunity according to Lombard et al. (2020).

Daily health assessments were conducted by a trained staff member to monitor both fecal and respiratory health. Fecal consistency was evaluated twice daily (before 0545 and 1600 h) from arrival to d 21 using a 0-to-3 point scale (0 = normal, firm but not hard; 1 = soft, piles but spreads slightly; 2 = runny, spreads easily; 3 = watery, splatters, devoid of solids; Renaud et al., 2020). Calves with a score of 0 or 1 were classified as normal, whereas those scoring 2 or 3 were considered abnormal (Renaud et al., 2020). A score of 3 was considered to be severe diarrhea. The duration of diarrhea was defined as the number of days with a fecal consistency score of 2 or 3. Because fecal consistency was assessed twice daily, the duration was calculated by summing the number of morning and afternoon observations with a scores ≥2 and divided by 2. Respiratory health assessments were conducted twice daily for the entire experiment, with a cumulative score assigned based on clinical signs: eye discharge (2 points), purulent nasal discharge (4 points), coughing (2 points), ear drooping or head tilt (5 points), and rapid or labored breathing (2 points; Love et al., 2014). If a calf's cumulative respiratory score reached 4, rectal temperature was evaluated (>39.5°C = 2 points). A score of ≥5 was classified as a respiratory event, indicative of BRD. The duration of respiratory illness was defined as the number of days with a score of ≥5, and was calculated by summing the number of morning and afternoon observations with a scores ≥5 and divided by 2.

### Treatment and Mortality Records

The treatment protocols were consistent across all groups and based on the scoring systems for either diarrhea or BRD. Calves exhibiting severe diarrhea (score of 3) received a single oral dose of meloxicam (meloxicam oral suspension; 3 mL/45 kg of BW; Solvet) at the first occurrence and were provided oral electrolytes until fecal consistency returned to normal (score 0 or 1). Electrolytes were prepared by dissolving 100 g of Truvitalyte (Truvital Animal Health, Palmerston, ON, Canada) in 2 L of warm water and offered via a nipple bottle. If calves refused to drink and showed signs of dehydration, electrolytes were administered using an esophageal tube. Electrolytes were also provided to the calf if milk was refused. If diarrhea persisted for 48 h after initial treatment or reoccurred, a second treatment was administered, consisting of 4 consecutive days of intramuscular injections of trimethoprim sulfadoxine (Borgal; 3.0 mL/45 kg of BW; Merck Animal Health), alongside an additional oral dose of meloxicam. In the rare circumstance that there were no improvements in fecal consistency 48 h following the 4 consecutive days, a third oral dose of meloxicam was administered alongside a subcutaneous injection of enrofloxacin (Baytril; 4.5 mL/45 kg of BW; Bayer Animal Health).

Treatment of respiratory disease was based on clinical presentation of disease (respiratory score ≥5). At the first signs of BRD, calves received subcutaneous injections of florfenicol (Nuflor; 6.0 mL/45 kg of BW; Merck Animal Health) and meloxicam (Metacam; 1.2 mL/45 kg of BW; Boehringer Ingelheim). If clinical signs reoccurred or persisted for 96 h, a second treatment included an intramuscular injection of marbofloxacin (Forcyl; 4.0 mL/45 kg of BW; Vetoquinol) along with an additional subcutaneous injection of meloxicam. In rare cases when respiratory disease persisted for 72 h after the second treatment or reoccurred, danofloxacin (A-180; 2.0 mL/45 kg of BW; Zoetis) was administered subcutaneously with an additional injection of meloxicam. All treatments were determined and administered by trained staff, with veterinary consultation sought in severe cases. Any treatments administered and mortalities were recorded.

### BW Measurements and Feed Efficiency Calculations

Following arrival to the facility, calves were weighed using a calibrated Tru-Test digital weigh scale (Tru-Test Datamars, Mineral Wells, TX) by farm staff before being moved into their individual pens. Calves were then weighed weekly (d 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, and 84) between 0800 and 1100 h. Average daily gain was calculated for each week by dividing the amount of weight gained per week (e.g., weight on d 7 − weight at arrival) by 7. To evaluate the DMI of the grain, grain intake was multiplied by the percentage of DM of either the starter (0.88) or corn blend (0.87), depending on which grain the calf received. Dry matter intake was then divided by 7 to estimate DMI on a daily basis. Feed-to-gain conversion was calculated weekly. To calculate the feed conversion ratio (**FCR**), the ME intake (Mcal) was determined using the individual liquid and solid feed intakes for each week (NRC, 2001; Quigley, 2007) and divided by the BW gain (kg) for that week.

### Sample Size and Power Calculation

A formal sample size calculation was not performed. The sample size was based on the records that were available from the calf-rearing facility.

### Statistical Analysis

All statistical analyses were performed using Stata 18 (StataCorp LP). Data were imported from Microsoft Excel (Microsoft Corp.) and assessed for completeness before analysis. Descriptive statistics were calculated for all variables. Differences between breeds in the Lombard categorization were evaluated using a Pearson chi-squared test. Except for DMI and FCR during the postweaning period, all observations were performed on individual calves. Calves were, however, housed in pairs during the postweaning period for 5 of the 10 groups (n = 320), and a fixed effect of pairing was included in the DMI and FCR models.

Cox proportional hazard and parametric survival models using a Weibull distribution were used to investigate time-to-event outcomes, including the first, second, and third diarrhea or respiratory treatments. Negative binomial regression models were built to evaluate count outcomes including the number of days with diarrhea or respiratory disease, number of antimicrobial treatments, and electrolyte doses. Mixed linear regression models with repeated measures were used to evaluate continuous variables including BW, ADG, DMI, and FCR. In all statistical models, group was used as a random effect to control for variations that may have existed between date of arrival or room, and breed (the variable of interest) was forced into each model.

The predictor variables of source (local dairy farm vs. auction) and arrival BW were screened for all models using univariable analyses with a liberal $P$-value ($P < 0.20$). Additionally, in the FCR model, morbidity was screened as a predictor variable. Variables meeting the criterion were considered for inclusion in multivariable models through a backward stepwise elimination process. Variables removed during this process were further evaluated for potential confounding, defined as a ≥20% change in the coefficient of the primary variable of interest. Interaction terms were assessed where biologically plausible; specifically, the interaction between breed and week was included in repeated measures linear regression models.

The Cox proportional hazards models were tested using the proportional hazards assumption with Shoenfeld residuals. In some cases, the assumptions were violated and a parametric survival model using Weibull distribution was used instead. Poisson models were originally built to investigate count outcomes; however, the likelihood ratio chi-squared tests indicated overdispersion and negative binomial regression models were fitted instead. Finally, the linear regression models were tested for the assumption of normality and heteroskedasticity, and the homogeneity of residuals was evaluated by graphing residuals against their predicted values and the distri-

**Table 2.** Distribution of STP concentrations at arrival to a calf-rearing facility by breed[1]

| STP category (g/dL; Lombard) | Holstein | | Crossbred | |
|---|---|---|---|---|
| | Number | Percent | Number | Percent |
| <5.1 | 107 | 29.5 | 35 | 23.5 |
| 5.1–5.7 | 92 | 25.3 | 44 | 29.5 |
| 5.8–6.1 | 39 | 10.7 | 16 | 10.7 |
| ≥6.2 | 125 | 34.4 | 54 | 36.2 |

[1]STP categorization included 512 of the 640 calves included in the study: Holstein (n = 363) and crossbred (n = 149).

bution of residuals was assessed using both scatter and quantile-quantile plots. Outliers were reviewed for biological plausibility and retained or removed accordingly. This process affected a total of 50 calves, each of which had between 1 and 3 data points removed for BW, ADG, DMI, or FCR. All final models met the assumptions of their respective analyses and demonstrated adequate fit.

## RESULTS

### Descriptive Statistics

A total of 640 calves were enrolled in the study; however, 26 calves (4.1%; Holstein n = 20; crossbred n = 6) had only health-scoring records and were therefore excluded from growth metrics and disease treatment analyses. Approximately half of all calves (47.8%) were sourced directly from farms, and the remaining came from auction facilities. Calves were received at the calf-rearing facility during 3 seasons: 192/640 (30%) in spring, 320/640 (50%) in summer, and 128/640 (20%) in fall. The average BW of all calves at arrival was 48.1 ± 3.7 kg (mean ± SD), with a range of 38.6 to 61.4 kg. Serum total protein concentrations were available for 512/640 (80%) of the calves upon arrival to the facility and the distribution across Lombard categories was found to be similar between breeds ($P = 0.54$; Table 2).

### Health Assessment

***Diarrhea.*** Among the 640 calves enrolled, 61% of crossbred calves (118/194) and 69% of Holstein calves (309/446) experienced at least 1 episode of diarrhea. Overall, the incidence of diarrhea was 23% higher (incidence rate ratio [**IRR**] = 1.23; 95% CI: 1.01 to 1.51; $P = 0.04$; Table 3] in Holstein compared with crossbred calves, which resulted in a greater number of days with diarrhea for Holsteins compared with crossbred calves. On average, Holsteins experienced 3.02 ± 0.37 d with diarrhea, and crossbred calves experienced 2.44 ± 0.34 d, accounting for the number of days visually scored.

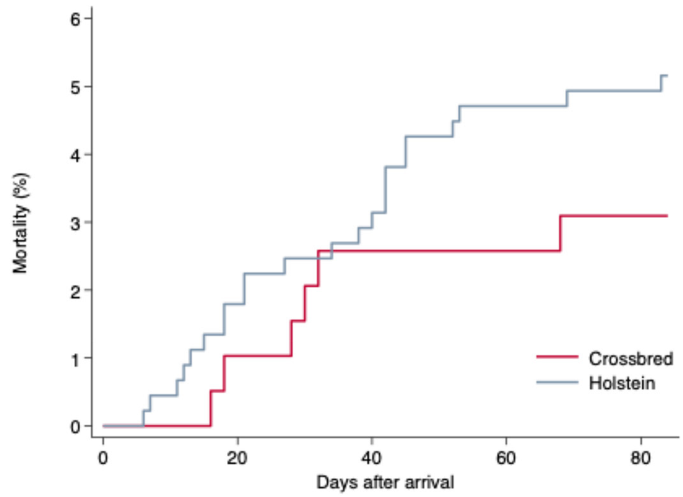


**Figure 1.** Kaplan–Meier failure curve for mortality after arrival of male Holstein (n = 446) and dairy × beef (n = 194) calves reared under similar conditions at a commercial calf-rearing facility.

Moreover, the incidence of severe diarrhea (score of 3) was 32% higher (IRR = 1.32; 95% CI: 1.06 to 1.64; $P = 0.01$; Table 3) in Holsteins compared with crossbreds, with Holsteins exhibiting severe diarrhea for an average of 2.79 ± 0.34 d versus 2.11 ± 0.30 d for crossbreds.

***Respiratory Disease.*** Overall, 270 calves (42.2%) had at least 1 respiratory score of ≥5 throughout the study period and were classified as having BRD. The number of days with a respiratory score ≥5 was not different between breeds (IRR = 1.39; 95% CI: 0.92 to 2.11; P = 0.12; Table 3).

***Treatment Intervention.*** A total of 62.9% (386/614) of calves were treated at least once for diarrhea. Of those treated, 32.9% (127/386) received only an nonsteroidal anti-inflammatory drug (**NSAID**), 61.6% (239/386) received both a first and second treatment with an NSAID and trimethoprim sulfadoxine, and 5.2% (20/386) received all 3 diarrhea treatments. Holstein calves tended to have a higher hazard for receiving the first NSAID treatment due to diarrhea (hazard ratio [**HR**] = 1.21; 95% CI: 0.97 to 1.52; $P = 0.09$) and had a higher hazard of receiving a single antibiotic treatment for diarrhea compared with crossbred calves (HR = 1.36; 95% CI: 1.03 to 1.79; $P = 0.03$; Table 3). No differences were detected between Holsteins and crossbred calves regarding the hazard of a second course of antibiotic therapy for diarrhea (HR = 1.45; 95% CI: 0.54 to 3.90; $P = 0.17$; Table 3).

Treatment of BRD was less prevalent, with 40.4% (248/614) of calves being treated 1 or more times for respiratory disease. Breed did not affect the hazard of receiving the first course of antibiotic therapy (HR = 1.08; 95% CI: 0.81 to 1.44; $P = 0.61$; Table 3). However,

**Table 3.** Effect of breed on health parameters of calves for an 84-d period at a calf-rearing facility

| Health parameter[1] | Breed | | Estimate (type)[2] | 95% CI | $P$-value[3] |
|---|---|---|---|---|---|
| | Holstein | Crossbred | | | |
| Mortality,[4] % (no.) | 5.2 (23/446) | 3.1 (6/194) | 1.63 (HR) | 0.66–4.03 | 0.286 |
| Diarrhea | | | | | |
| Days with diarrhea | 3.0 | 2.4 | 1.23 (IRR) | 1.01–1.51 | 0.041 |
| Days with severe diarrhea | 2.79 | 2.11 | 1.32 (IRR) | 1.06–1.64 | 0.013 |
| 1st diarrhea treatment,[5] % (no.) | 64.8 (276/426) | 57.4 (108/188) | 1.21 (HR) | 0.97–1.52 | 0.089 |
| 2nd diarrhea treatment,[5] % (no.) | 45.5 (194/426) | 35.6 (67/188) | 1.36 (HR) | 1.03–1.79 | 0.032 |
| 3rd diarrhea treatment,[5] % (no.) | 3.3 (14/426) | 3.2 (6/188) | 1.45 (HR) | 0.54–3.90 | 0.460 |
| Doses of electrolytes[5] | 7.4 | 6.2 | 1.18 (IRR) | 0.93–1.50 | 0.173 |
| Respiratory | | | | | |
| Respiratory disease, % (no.) | 43.5 (194/446) | 39.2 (76/194) | 1.13 (OR) | 0.79–1.63 | 0.486 |
| Days with respiratory disease | 2.1 | 1.5 | 1.39 (IRR) | 0.92–2.11 | 0.122 |
| 1st respiratory treatment,[5] % (no.) | 40.4 (172/426) | 36.2 (68/188) | 1.08 (HR) | 0.81–1.44 | 0.608 |
| 2nd respiratory treatment,[5] % (no.) | 23.0 (98/426) | 13.8 (26/188) | 1.61 (HR) | 1.04–2.51 | 0.032 |
| 3rd respiratory treatment,[5] % (no.) | 12.2 (52/426) | 5.3 (10/188) | 2.47 (HR) | 1.24–4.91 | 0.010 |

[1]Parameters related to diarrhea occurred within the first 21 d at the facility, and respiratory parameters were evaluated over the entire experimental period.

[2]OR = odds ratio, HR = hazard ratio, and IRR = incidence rate ratio. Ratios were estimated using multilevel logistic regression, Cox proportional hazards model, and multilevel negative binomial regression (adjusted for overdispersion) in Stata, respectively. The OR were used for binary outcomes (e.g., diarrhea positive = 1), HR for time-to-event data (e.g., time to first diarrhea treatment), and IRR for count data (e.g., number of days with diarrhea).

[3]$P$-value of comparison between breeds. Significant differences: $P \leq 0.05$, tendencies $0.05 < P \leq 0.10$.

[4]Mortality and morbidity parameters include a total of 640 calves: Holstein (n = 446) and crossbred (n = 194).

[5]Treatment parameters include a total of 614 calves: Holstein (n = 426) and crossbred (n = 188).

Holstein calves had a higher hazard of receiving a second (HR = 1.62; 95% CI: 1.04 to 2.51; $P = 0.03$; Table 3) or third course of antibiotic therapy than crossbred calves (HR = 2.47; 95% CI: 1.24 to 4.91; $P = 0.01$; Table 3).

***Mortality.*** Twenty-nine (4.5%) calves died within the 84 d at the rearing facility, of which Holsteins accounted for 79% (23/29); the remaining 21% were crossbred calves (6/29). A Kaplan–Meier failure function was used to visually assess the cumulative incidence of mortality after arrival to the facility, stratified by breed (Figure 1). Although numerically a greater proportion of Holsteins died, there was no difference in the hazard for mortality between breeds (HR = 1.63; 95% CI: 0.66 to 4.03; $P = 0.29$; Table 3).

### Growth Performance

Overall, BW was affected by breed ($P < 0.01$; Figure 2a) and an interaction occurred between breed and week ($P < 0.01$). Upon arrival, Holstein calves were 0.3 ± 0.12 kg heavier than crossbred calves (95% CI: 0.03 to 0.53 kg; $P = 0.02$); however, from d 7 to 28, the difference in BW between breeds was not significant ($P > 0.05$). At d 28, crossbred calves weighed 1.0 ± 0.48 kg more than Holsteins (95% CI: 0.04 to 1.93 kg; $P = 0.04$), and on d 35 the crossbred calves weighed 1.8 ± 0.57 kg more than Holsteins (95% CI: 0.71 to 2.93 kg; $P < 0.01$). On the first day of weaning (d 42), crossbred calves weighed 2.9 ± 0.70 kg (95% CI: 1.58 to 4.31 kg; $P < 0.01$) more than Holsteins, and 4.4 ± 0.95 kg more (95% CI: 2.52 to 6.26 kg; $P < 0.01$) by d 56. On the final day of weaning (d 63), this difference increased to 5.5 ± 1.09 kg (95% CI: 3.33 to 7.61 kg; $P < 0.01$). This continued throughout the remainder of the study: crossbred calves weighed 6.5 ± 1.25 kg (95% CI: 4.00 to 8.90 kg; $P < 0.01$) and 7.1 ± 1.31 kg (95% CI: 4.56 to 9.70 kg; $P < 0.01$) more than Holsteins on d 77 and 84, respectively.

Similarly, ADG was significantly higher in crossbred calves compared with Holsteins ($P < 0.01$; Figure 2b) and there was a breed by week interaction ($P < 0.01$). Between arrival and d 7, crossbred calves tended to gain 0.1 ± 0.03 kg/d more than Holsteins (95% CI: −0.01 to 0.12 kg/d; $P = 0.10$). However, ADG was similar between breeds between d 7 and 28 ($P > 0.10$). Differences in ADG were detected between d 28 and 35 (0.1 ± 0.03 kg/d; 95% CI: 0.06 to 0.19 kg/d; $P < 0.01$), d 35 to 42 (0.1 ± 0.03 kg/d; 95% CI: 0.08 to 0.21 kg/d; $P < 0.01$), and d 42 to 49 (0.2 ± 0.04 kg/d; 95% CI: 0.08 to 0.22 kg/d; $P < 0.01$), where crossbred calves had higher ADG than Holsteins. Between d 56 to 63, when calves were undergoing their final step down of weaning, crossbred calves gained 0.2 ± 0.05 kg/d more than Holsteins (95% CI: 0.09 to 0.28 kg/d; $P < 0.01$). No differences occurred in ADG from d 63 to 70 or 70 to 77 ($P = 0.16$ and $P = 0.29$, respectively), however, from d 77 to 84 crossbred calves began to outperform the Holsteins again, gaining 0.2 ± 0.05 kg/d more than Holsteins (95% CI: 0.06 to 0.26 kg/d; $P < 0.01$).

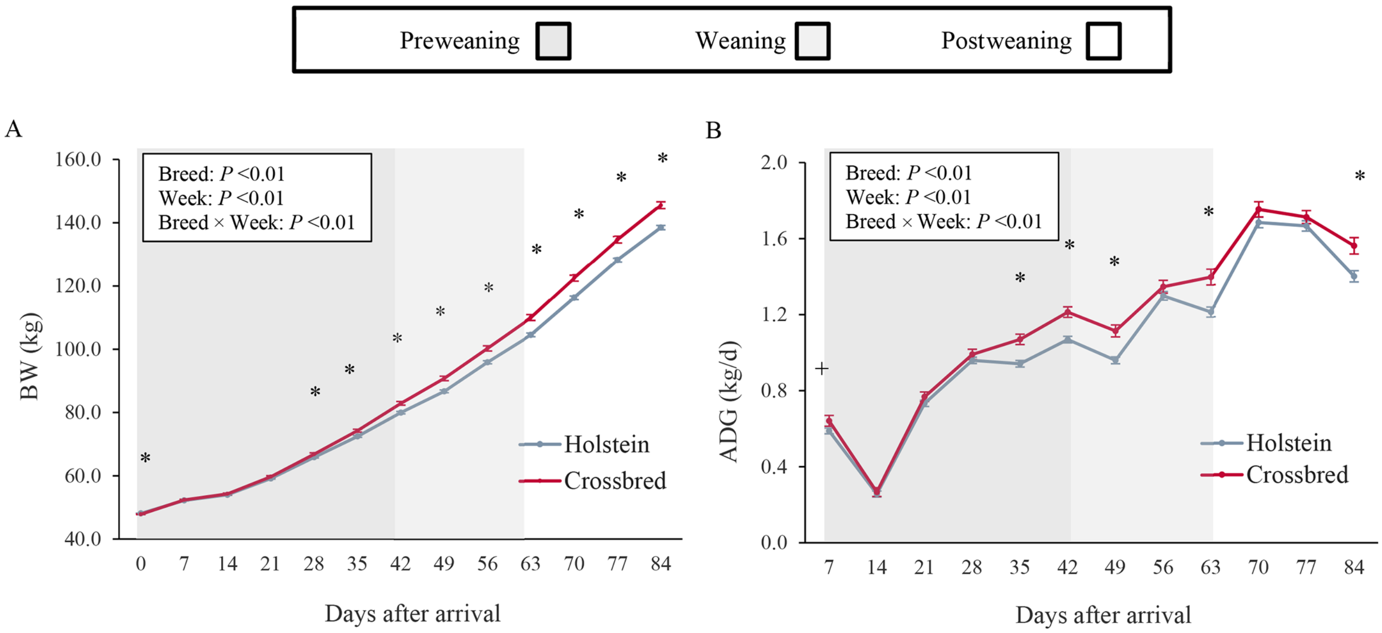


**Figure 2.** The effect of breed on growth performance of (A) BW and (B) ADG for d 0 to 84. Male Holstein (n = 426) and dairy × beef (n = 188) calves were raised at a single calf-rearing facility and managed under similar conditions. Error bars denote SEM. Significant differences: $*P \le 0.05$, tendencies $+ \; 0.05 < P \le 0.10$.

From arrival until d 28, starter intake was minimal and no significant differences in DMI were observed between the breeds ($P > 0.05$; Figure 3). However, between d 28 and 35, crossbred calves consumed 0.06 ± 0.03 kg/d more starter on a DM basis than Holsteins (95% CI: 0.01 to 0.12 kg/d; $P = 0.02$). Differences in DMI continued, with crossbreds consuming 0.10 ± 0.04 kg/d (95% CI: 0.02 to 0.17 kg/d; $P = 0.01$), 0.12 ± 0.06 kg/d (95% CI: 0.00 to 0.24 kg/d; $P = 0.04$), and 0.13 ± 0.05 kg/d (95% CI: 0.03 to 0.22 kg/d; $P = 0.01$) more grain than Holsteins from d 35 to 42, 42 to 49, and 49 to 56, respectively. Following weaning, there were no differences in DMI between Holsteins and crossbreds ($P = 0.45$).

Finally, differences in FCR were demonstrated between breeds throughout the rearing phase ($P < 0.05$; Figure 4). Within the first week at the rearing facility, Holsteins had a tendency to require 0.60 ± 0.32 Mcal/kg more energy than crossbreds to achieve the same weight gain (95% CI: −0.02 to 1.22 Mcal/kg; $P = 0.06$). Differences in FCR also occurred from d 28 to 35 and 35 to 42, as well as during weaning on d 42 to 49, where crossbred calves required fewer Mcal of energy per kilogram of BW gain compared with Holsteins ($P < 0.05$). In the postweaning period, differences in FCR between breeds were minimal, although from d 70 to 77, Holstein calves had improved FCR compared with crossbreds and required 1.20 ± 0.47 Mcal/kg less than crossbreds (95% CI: 0.28 to 2.13 Mcal/kg; $P = 0.01$). However, this finding did not persist into the final week at the rearing facility (95% CI: −0.56 to 1.53 Mcal/kg; $P = 0.36$).

## DISCUSSION

This study described the association of breed with morbidity, mortality, and growth performance in Holstein and dairy × beef calves. Overall, our results demonstrated that crossbred calves had improved health outcomes compared with Holstein calves and that they outperformed Holsteins, specifically attributed to improvements in BW, ADG, DMI, and FCR. These findings emphasize the benefits of raising crossbred calves over Holsteins at calf-rearing facilities.

The health and performance of a calf are strongly influenced by their ability to cope with stressors, both environmental and physiological. In the current study, we found that incidence of diarrhea was lower for crossbred calves than for Holstein calves. Similarly, a recent study reported a lower odds of diarrhea for dairy × beef crossbred calves compared with Holsteins (Moroz et al., 2025). These findings suggest that crossbreeding provides an opportunity to improve the resilience of the calf to health challenges through heterosis or hybrid vigor (Sørensen et al., 2008). A reduction in the duration of diarrhea is advantageous because diarrhea has both short- and long-term consequences that negatively affect the health, welfare, and productivity of an animal (Pardon et al., 2013). Studies have indicated that diarrheic calves present sickness behaviors, including a reduction in milk consumption and activity level, increasing the risk of dehydration, and ultimately the risk of mortality (Guevara-Mann et

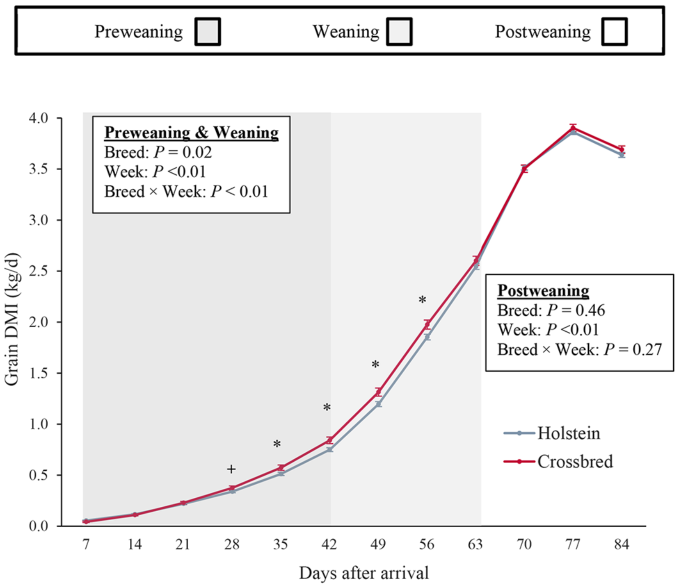


**Figure 3.** Average daily DMI of grain-fed ad libitum to Holstein (n = 426) and dairy × beef (n = 188) calves. Weekly intake data were divided by 7 to estimate daily intakes. Error bars denote SEM. Significant differences between breeds: $*P \leq 0.05$, tendencies between breeds $+\ 0.05 < P \leq 0.10$.

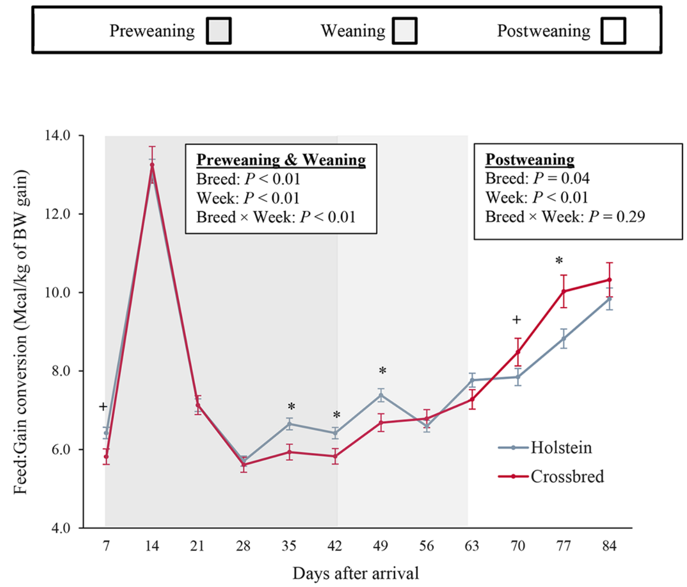


**Figure 4.** Feed-to-gain conversion ratios measured in Holstein (n = 426) and dairy × beef (n = 188) calves raised at a single calf-rearing facility and managed under similar conditions. Error bars denote SEM. Significant differences: $*P \leq 0.05$, tendencies $+\ 0.05 < P \leq 0.10$.

al., 2023). Additionally, prolonged diarrhea is associated with reduced BW and ADG throughout the rearing phase (Schinwald et al., 2022). Therefore, the findings of this study highlight the benefits of crossbreeding on both the health and welfare of the calf, while potentially reducing the negative effects on growth performance.

Despite the differences in the incidence of diarrhea, we did not observe differences in the incidence of respiratory disease between breeds, which aligns with previous findings (Moroz et al., 2025). However, in the present study, fewer crossbred calves received a second or third antimicrobial treatment for respiratory disease. Reducing the use of antimicrobial in the calf-rearing industry is imperative for improving public trust and slowing the development of resistant pathogens that can compromise both animal and human health (Uyama et al., 2022). The findings of this study suggest that crossbred calves have improved recovery from respiratory disease compared with Holsteins, which could offer a straightforward and efficient approach to improving antimicrobial stewardship in calf rearing. Similarly, Fernandes et al. (2025) highlighted Holstien × Angus crossbred calves' ability to recover from pneumonia without the use of antimicrobials, compared with Holsteins. Reducing the industry's reliance on antimicrobials is not only important for protecting future vaccine efficacy, but also has important economic implications, beginning with treatment costs (Stanton et al., 2012). Dubrovsky et al. (2020) reported that preweaning calves treated for respiratory disease one or more times incurred additional costs between $31.57 and $36.95 USD per treatment. These costs reflected drug and labor expenses, along with losses in ADG, underscoring the negative economic impact of increased antimicrobial use. Crossbreeding's ability to reduce the days with diarrhea, paired with fewer antimicrobial interventions for disease recovery, presents an attractive combination of attributes that may encourage producers to raise these heartier and more resilient young calves.

Breed-related differences extend beyond health, affecting the growth performance of calves as well. To ensure the successful transition of the young calf to a functioning ruminant, it is critical that the calf consumes adequate amounts of the solid diet during the preweaning and weaning periods (Baldwin et al., 2004; Coverdale et al., 2004). The solid diet stimulates the development of the rumen by promoting the growth of rumen papillae, which in turn increases the surface area available for nutrient absorption, while also increasing microbial fermentation (Baldwin et al., 2004). Moreover, starter intake before and during the weaning process is essential for long-term productivity. Calves that consume more starter during these periods have been shown to have higher intakes and improved weight gain following weaning (Khan et al., 2011). In the current study, we found that DMI of the solid diet was lower for Holsteins compared with crossbred calves. The difference began around d 35 and persisted during the first weeks of weaning, suggesting that crossbred calves were more prepared for the complete transition from milk to solid feed. Ware et al. (2015) found a similar trend, reporting that Holstein × Jersey calves had greater DMI than the average of the purebred calves. The greater intakes observed before and during weaning in crossbred calves likely contributed to their increased ADG and heavier BW compared with Holsteins. However, the differences in growth performance should be attributed not only to greater intakes of the solid diet, but also heterosis.

Increased inbreeding in the dairy industry has previously been associated with reduced calfhood survival and slower growth during the preweaning period (McParland et al., 2008; Davis and Simmen, 2010). The greater growth performance and improved FCR observed in the crossbred calves in this study suggest that crossbreeding could help mitigate these negative effects by introducing greater genetic diversity, consistent with previous findings (Berry, 2021; Moroz et al., 2025). Given that beef breeds are characterized by rapid and efficient growth and dairy breeds have been genetically selected for high milk production, this incorporation of beef genetics in the offspring could partially explain the performance differences observed between the 2 breeds in this study (Wathes et al., 2014). These improvements in efficiency can then translate to economic benefits for the producer. For instance, Arens et al. (2023) reported that crossbred calves (Montbéliarde × Limousine and Viking Red × Limousine) had lower cost per kilogram of gain compared with Holstein calves in the preweaning period. To further understand the role of genetics in early-life growth, future research should consider raising purebred Holstein, Holstein × Angus, and purebred Angus calves, from embryos implanted into Holsteins cows, under standardized rearing conditions.

Some limitations must be considered when interpreting the results of this study. First, when relying on clinical scoring systems to monitor health outcomes in Holstein and crossbred calves, some degree of visual bias is unavoidable. It is also possible that the 2 breeds exhibit clinical signs of illness differently. Although measures were implemented to minimize this bias and ensure that treatment decisions were applied consistently across breeds, this remains a limitation of the study. Moreover, the age of calves upon arrival to the rearing facility was unknown. Although North American transportation regulations set minimum age requirements, previous research has found that most male calves in Canada are sold within the first 10 d of life (Wilson et al., 2023). Based on this, we estimated that the calves in the present study arrived at the rearing facility at less than 10 d of age. Furthermore, during the postweaning phase, feed intake from paired calves (320/640) was measured at the pen level and allocated equally to each calf for analysis. This approach assumed uniform intake within a pair, and was implemented as a convenience, as individual intakes were not obtained. Because pairing was not based on breed, the measurements for DMI and FCR during the postweaning period may not accurately reflect individual variance in intake and may explain the limited differences during this period. Pair housing has also been shown to alter feeding behaviors of calves, increasing solid feed intake and subsequently weight gain (Costa et al., 2016). Finally, an imbalance existed in sample size between the 2 breeds included in this study, which may have influenced the interpretation of breed effects.

## CONCLUSIONS

These results demonstrate that dairy × beef crossbred calves experienced a lower incidence of diarrhea, received fewer antimicrobial treatments, and exhibited improved growth performance compared with Holstein calves during the rearing period. Crossbreeding may offer producers a practical strategy to enhance calf resilience and production efficiency. Because all calves in this study were managed under similar environmental conditions, future research should explore how varying management practices and environmental challenges may influence the ability of different breed types to cope with stressors while maintaining performance.

## NOTES

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**Nonstandard abbreviations used:** BRD = bovine respiratory disease; FCR = feed conversion ratio; HR = hazard ratio; IRR = incidence rate ratio; MR = milk replacer; NSAID = nonsteroidal anti-inflammatory drug; OR = odds ratio; STP = serum total protein.

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